

ETHAN SHARP

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EDUCATION

The University of Tennessee, Knoxville

Bachelor of Science in Biomedical Engineering with a Minor in Applied Music

Graduation: May 2027

Cumulative GPA: 3.56/4.00

EXPERIENCE

University of Tennessee, Department of Chemical and Biomolecular Engineering

Knoxville, TN

Undergraduate Research Assistant

August 2026 – Present

- Collaborating with Tracy Ann Bruce-Tagoe in the lab of Dr. Danquah to investigate RNA aptamers as an antibody substitution
- Gaining knowledge and experience in the process of aptamer development and testing as well as DNA and RNA synthesis
- Performing experimentation using a surface plasmon resonance machine to detect aptamer-protein interaction

Knox County Democratic Party

Knoxville, TN

Field Organizer

May 2026 – Present

- Conducting door-to-door canvassing to engage with hundreds of Knox County residents and drive voter turnout for local elections
- Researching and cataloging high-profile donors into structured databases to streamline outreach and phone banking campaigns
- Managing data in VoteBuilder by segmenting voters by district and political affinity to better create targeted canvassing routes

University of Tennessee, Knoxville

Knoxville, TN

Office Assistant

August 2024 – Present

- Managing logistics and package distribution for 1,660+ residents using S.C. Logic and Microsoft Excel
- Auditing and maintaining digital records for key and card access
- Monitoring data from the facility fire panel, coordinating rapid-response dispatch from maintenance to investigate alarms

University of Tennessee, Department of Chemical and Biomolecular Engineering

Knoxville, TN

Undergraduate Research Assistant

May 2026 – August 2026

- Collaborated with Dr. Paul Dalhaimer to investigate the impact of PEG-PLGA nanoparticle geometry on targeted drug delivery
- Gained hands-on laboratory experience maintaining cell cultures, performing experimental assays, and managing sterile procedures in a biosafety cabinet
- Executed analytical testing and imaging protocols including Western blots, cell sample collection, microscopy, and infrared imaging

RELEVANT PROJECTS

The University of Tennessee, Knoxville

Knoxville, TN

The Utilization of LCHN Technology in Combination with pH Reactive PLGA Core Synthesis for the Treatment of Acute Myeloid Leukemia

March – May 2026

- Proposed a biomimetic LCHN delivery platform with pH-sensitive PLGA cores to target AML cells and repair bone marrow
- Designed a modular payload releasing Daunorubicin to eradicate AML and G-CSF with ECM peptides for hematopoietic recovery
- Formulated an experimental framework outlining double-emulsion synthesis, multi-instrument validation, and in vivo murine testing

Dual Blood Glucose Monitor and Insulin Pump for Type 1 Diabetics

May 2025

- Designed an integrated continuous glucose monitor and automated insulin pump for patients with Type 1 Diabetes Mellitus
- Analyzed the physiological impacts of hyperglycemia and ketoacidosis to define sensor-to-actuator feedback requirements
- Addressed the need for automated blood glucose regulation to prevent organ damage caused by chronic insulin deficiency

Telescoping Rod with Bioactive Glass Modification Proposal

November – December 2024

- Collaborated with a 5-person engineering team to develop a technical proposal for modifying pediatric telescoping rods
- Co-engineered a design utilizing a bioactive glass surface layer to promote osteoblast attachment and improve the long-term osseointegration of stainless steel and titanium implants
- Designed testing protocols to validate the modification, including pull-out tests for mechanical stability and resazurin assays for biocompatibility and cell viability

Autonomous Sphero Rover Project

November 2023

- Developed MATLAB code to autonomously navigate a Sphero rover to user-selected coordinates on a mapped interface
- Programmed a "detect and avoid" logic using distance sensors to pause rover movement when obstacles were present
- Integrated real-time data visualization by graphing the rover's position to track progress toward the destination

SKILLS & CERTIFICATIONS

Laboratory & Analytical

Mammalian Cell Culture, Biosafety Cabinet Sterile Technique, Western Blotting, Infrared/Microscopy Imaging, Experimental Design, SPR testing

Software and Programming

Python, JavaScript, MATLAB, HTML, R, SolidWorks, CSS, Onshape, VoteBuilder

Technical & Fabrication

Manual machining, mechanical assembly, prototyping, mechanical testing,